

Remasking Strategies in MDMs: Empirical Analysis and Research Directions

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1 Overview

This document is a companion to `docs/comparison.md` (literature review). Its purpose is twofold:

1. **Empirical analysis** — interpret the step-sweep results ($T \in \{128, 256, 512, 1000\}$) through the statistical framework built in the literature review.
2. **Research directions** — propose concrete, formally grounded extensions of the work.

All results come from evaluations on Bocconi HPC A100 80GB (N=100 samples, seed=42, OWT reference corpus of 1000 samples). Raw numbers are in `results/combined_comparison.md`.

2 Statistical Framework: Connecting Metrics to Theory

2.1 The Three Metrics and What They Actually Measure

Understanding what each metric captures statistically is prerequisite to interpreting results.

2.1.1 Generation Perplexity (gen_ppl)

$$\text{gen_ppl} = \exp \left(\frac{1}{N} \sum_{n=1}^N \frac{1}{|x^{(n)}|} \sum_{i=1}^{|x^{(n)}|} -\log p_{\text{GPT-2}}(x_i^{(n)} \mid x_{<i}^{(n)}) \right)$$

What it measures: The exponential of the average per-token cross-entropy assigned by GPT-2 large to the generated texts. Formally, gen_ppl is an empirical estimate of:

$$\text{gen_ppl} \approx \exp(H(p_\theta, q_{\text{GPT2}}))$$

where $H(p_\theta, q_{\text{GPT2}}) = \mathbb{E}_{x \sim p_\theta}[-\log q_{\text{GPT2}}(x)]$ is the **cross-entropy** of GPT-2’s distribution q with respect to the model’s generated distribution p_θ .

Important decomposition. Cross-entropy decomposes as:

$$H(p_\theta, q_{\text{GPT2}}) = \underbrace{H(p_\theta)}_{\text{entropy of generator}} + \underbrace{\text{KL}(p_\theta \parallel q_{\text{GPT2}})}_{\text{distributional gap to GPT-2}}$$

gen_ppl therefore measures both the *entropy* of the generated distribution and the *KL divergence* from GPT-2. It is **not** a pure KL divergence. A model that generates highly diverse text (high $H(p_\theta)$) will have higher gen_ppl than a mode-collapsing model (low $H(p_\theta)$) even if both are equally similar to GPT-2. This is the fundamental reason gen_ppl and MAUVE (which rewards diversity) can diverge.

Statistical interpretation: - gen_ppl penalises both **distributional gap** (KL term) and **lack of concentration** (entropy term). Mode-collapsing generators suppress $H(p_\theta)$, thereby reducing gen_ppl *even if* their KL to GPT-2 is unchanged. This makes gen_ppl a misleading quality metric whenever diversity is also valued. - It is biased towards favouring text that is GPT-2-like (low cross-entropy with GPT-2’s training data), regardless of coverage of the true reference distribution. - A model that generates a single high-quality sentence repeated 100 times will achieve very low gen_ppl but be useless. This is the **mode collapse failure mode**, where p_θ concentrates on a single mode of q_{GPT2} . - gen_ppl is a **point estimate** with high variance at $N=100$. The empirical standard error of the mean is approximately $\hat{\sigma}/\sqrt{N}$ where $\hat{\sigma}$ is the sample standard deviation of per-sequence cross-entropies.

What it does NOT measure: - Diversity or coverage of the text distribution. - Whether the generated texts are factually correct or semantically coherent. - Alignment with the model’s actual training distribution (OWT), only with GPT-2’s.

2.1.2 MAUVE

MAUVE (Pillutla et al., 2021) measures the divergence between the generated distribution p_θ and a reference corpus q_{ref} using a soft variant of the Kullback–Leibler divergence. The computation proceeds as follows:

1. Embed all generated texts and all reference texts using a large LM (GPT-2 XL by default).
2. In the resulting embedding space, fit a single Gaussian mixture model (GMM) with k components to the union of both sets.
3. Compute the “soft” precision and recall: for each component, what fraction of probability mass belongs to generated vs. reference? The resulting frontier traces out a Precision–Recall curve analogous to GANs.
4. MAUVE is the area under this curve, normalised to $[0, 1]$.

Formal definition. Let P (generated) and Q (reference) be empirical distributions over sequences projected via a fixed LM encoder. Define the mixture:

$$M_\lambda = (1 - \lambda)P + \lambda Q, \quad \lambda \in [0, 1]$$

The **divergence frontier** is the parametric curve:

$$\mathcal{F}(P, Q) = \{(\text{KL}(M_\lambda \| Q), \text{KL}(M_\lambda \| P)) : \lambda \in [0, 1]\} \subset \mathbb{R}_{\geq 0}^2$$

At $\lambda = 0$: $M_0 = P$, giving the point $(\text{KL}(P \| Q), 0)$ — measures how unlike the reference the generated distribution is (precision failure). At $\lambda = 1$: $M_1 = Q$, giving the point $(0, \text{KL}(Q \| P))$ — measures how much of the reference is not covered (recall failure). The frontier traces a monotone curve between these extremes. MAUVE is:

$$\text{MAUVE}(P, Q) = \exp(-c \cdot \text{Area}(\mathcal{F})) \in [0, 1]$$

where $\text{Area}(\mathcal{F})$ is the area enclosed between the frontier curve and the two axes, and $c > 0$ is a fixed constant. A frontier close to the origin (both KL terms small) gives $\text{Area} \approx 0$ and $\text{MAUVE} \approx 1$.

In practice, the population distributions P and Q are approximated by empirical samples, the LM encoder is GPT-2 XL (producing 1600-dimensional representations), and the KL terms are estimated via a k -component Gaussian mixture model fit to the union of both sample sets. This quantisation introduces additional estimation variance beyond the finite-sample variance of P and Q themselves.

The **double-sided** structure means MAUVE penalises both **precision failure** (generating text unlike the reference, $\text{KL}(P \| Q)$ large) and **recall failure** (failing to cover the reference, $\text{KL}(Q \| P)$ large).

Statistical interpretation: - MAUVE is a **two-sided** distributional divergence: it captures both precision and recall. A mode-collapsing model loses recall (fails to cover Q) and gets a low MAUVE score even if individual samples score high on `gen_ppl`. - MAUVE is **sensitive to N**: with $N=100$ samples, the empirical P is a coarse approximation of p_θ . The 1600-dimensional embedding space means the GMM is fitted on a sparse cloud (100 points in 1600 dimensions), and the KL estimates from the GMM have high variance. Pillutla et al. (2021) demonstrate that MAUVE variance decreases substantially as N increases from 100 to 500; beyond $N=500$, estimates stabilise for their benchmarks. Our $N=100$ estimates should be treated as **directional indicators** of relative ordering between strategies, not precise absolute values. Confidence intervals (obtainable via bootstrap resampling of the generated texts) are not reported here but are needed before claiming statistical significance. - MAUVE is **reference-dependent**: the value is only meaningful relative to a fixed reference corpus. Our OWT reference (1000 samples, `skip=5000`) differs from the paper’s reference, making absolute comparisons across papers meaningless. Only **within-experiment** comparisons (same reference, same N) are valid.

Why gen_ppl and MAUVE can diverge. From the cross-entropy decomposition:

$$H(p_\theta, q) = H(p_\theta) + \text{KL}(p_\theta \| q)$$

Minimising gen_ppl is equivalent to minimising $H(p_\theta) + \text{KL}(p_\theta \| q)$. A mode-collapsing generator suppresses $H(p_\theta)$ while possibly also suppressing $\text{KL}(p_\theta \| q)$ — lower gen_ppl but at the cost of diversity. MAUVE, being symmetric in P and Q via the frontier, penalises coverage failure and thus rewards diversity. These are **genuinely competing objectives**: any strategy that reduces gen_ppl by reducing $H(p_\theta)$ (rather than by reducing the KL term) will simultaneously hurt MAUVE. This tension is the statistical explanation for the inversion we observe as T increases.

2.1.3 Entropy

$$H = - \sum_c \frac{n_c}{N_{\text{total}}} \log_2 \frac{n_c}{N_{\text{total}}}$$

where n_c is the count of character c across all generated texts. This is character-level marginal entropy.

Statistical interpretation: - Character entropy is a **weak lower bound** on text diversity: it captures the marginal distribution of characters, not their joint distribution or semantic diversity. - A system that generates grammatically correct but semantically repetitive text can maintain high character entropy (many different characters, words) while exhibiting low semantic diversity. - We use entropy as a **diversity proxy**, not a direct measure of generation quality. A meaningful drop in entropy (e.g. > 0.1 bits) across strategies is evidence of mode collapse or reduced vocabulary coverage.

Relationship to MAUVE. Entropy and MAUVE both capture diversity, but at different scales: entropy operates on the marginal token distribution, while MAUVE operates on the joint sentence embedding distribution. A drop in entropy without a drop in MAUVE would suggest reduced character-level diversity without distributional shift. A drop in MAUVE without a drop in entropy would suggest distributional shift without character-level change — sentences using similar characters but in less natural combinations.

2.2 The Factorization Error Framework

All three metrics can be grounded in the **factorization error** framework introduced in the thesis (following Lavenant & Zanella, 2024 and EB-Sampler):

$$E_{\text{fact}}(T) = \mathbb{E} \left[\text{KL} \left(p_\theta(\cdot | x_{\mathcal{M}}) \parallel \prod_{i \in \mathcal{M}} p_\theta(x^i | x_i) \right) \right]$$

where $\mathcal{M}$ is the set of simultaneously unmasked positions. This error arises because the model predicts each masked position independently (product factorisation) while the true joint distribution has dependencies.

Connection to our metrics: - **gen_ppl** correlates with E_{fact} through the marginal quality of individual tokens. Lower $E_{\text{fact}} \rightarrow$ better conditioning $\rightarrow$ lower cross-entropy per token. - **MAUVE**

is more sensitive to the *joint* distribution of generated text. If E_{fact} causes strong tokens to be over-committed early (as in remdm-conf), the resulting text may have good marginal token quality (low gen_ppl) but poor joint coherence (low MAUVE). - **Entropy** measures the marginal diversity of the generated distribution, which is degraded when E_{fact} causes systematic early token commitment (mode-seeking behaviour).

Key insight: Remasking is a method for reducing E_{fact} at inference time. Different remasking strategies trade off the three metrics differently because they reduce E_{fact} in different ways and with different side effects on diversity.

3 Empirical Results: A Statistical Reading

3.1 Complete Results Table

All results: N=100 samples, seed=42, OWT reference (1000 samples, skip=5000), evaluated on Bocconi HPC A100 80GB.

Generation Perplexity (gen_ppl) — lower is better:

Strategy	T=128	T=256	T=512	T=1000	$\Delta(128 \rightarrow 1000)$
mdlm	60.914	54.202	49.019	52.269	−14.2%
remdm-conf	57.579	50.668	42.868	37.321	− 35.2%
remdm-loop	59.632	42.877	34.322	30.296	− 49.2%

MAUVE — higher is better:

Strategy	T=128	T=256	T=512	T=1000	$\Delta(128 \rightarrow 1000)$
mdlm	0.170	0.740	0.592	0.590	+247%
remdm-conf	0.440	0.475	0.470	0.325	−26%
remdm-loop	0.396	0.614	0.532	0.684	+73%

Character-level entropy (H, bits) — proxy for diversity:

Strategy	T=128	T=256	T=512	T=1000	$\Delta(128 \rightarrow 1000)$
mdlm	5.507	5.481	5.440	5.446	−0.061
remdm-conf	5.499	5.443	5.405	5.357	− 0.142
remdm-loop	5.538	5.460	5.427	5.390	−0.148

3.2 Finding 1: gen_ppl and MAUVE Decouple — Diversity Cost of Remasking

The cleanest statistical finding is the **gen_ppl–MAUVE decoupling** for remdm-conf at high T:

- remdm-conf achieves the best gen_ppl among all strategies at T=1000 (37.3), yet the worst MAUVE (0.325). remdm-loop achieves the worst gen_ppl at T=128 (59.6) yet eventually the best MAUVE at T=1000 (0.684).

The directional pattern (remdm-conf gen_ppl improving, remdm-conf MAUVE declining, remdm-loop MAUVE improving) is consistent across all four step counts $T \in \{128, 256, 512, 1000\}$, which is evidence against noise as the sole explanation. However, without bootstrap confidence intervals, the precise magnitudes should be treated as indicative only.

Mechanistic explanation. remdm-conf applies the standard MDLM unmasking step and then, as a *remasking* correction, re-masks committed tokens with probability $\sigma_t \cdot (1 - c^i)$, where $c^i = H(p_\theta(x^i | z_t))$ is the confidence. Tokens with high confidence (low entropy) survive remasking with high probability. This is a **greedy remasking policy**: it systematically protects high-confidence committed tokens from being revisited.

As T increases, each individual denoising step uncovers fewer masked positions (finer step size), so the context available when each token is first committed becomes progressively richer. Richer context $\rightarrow$ lower entropy $\rightarrow$ higher confidence $\rightarrow$ lower remasking probability. Tokens committed at high T steps are therefore almost never remasked even if the commitment was locally correct but globally inconsistent. The result is a sequence that is locally coherent (each token well-predicted by its context $\rightarrow$ low `gen_ppl` from the cross-entropy decomposition’s KL term) but globally mode-seeking (few diverse trajectories were explored $\rightarrow$ reduced $H(p_\theta) \rightarrow$ MAUVE recall failure).

This is the **mode-seeking vs mode-covering** trade-off (Theis et al., 2016): greedy remasking incentivises mode-seeking (preserve the most likely token at each step), while loop remasking is more mode-covering (revisit all positions, allowing less confident but globally coherent trajectories). The `gen_ppl`–MAUVE decoupling is the empirical signature of this trade-off.

Practical consequence. If the downstream use case requires fluent text (`gen_ppl` proxy), `remdm-conf` is preferred at high T . If the use case requires diverse, reference-matching text (MAUVE proxy, e.g. training data generation), `remdm-loop` dominates at $T = 512$.

3.3 Finding 2: The MDLM Diversity Window at $T=256$

MDLM MAUVE peaks at $T=256$ (0.740) — higher than either remasking strategy at that step count — then drops to ~ 0.59 and plateaus. This is the strongest MAUVE score in our entire experiment, achieved by the baseline model with no remasking.

Statistical interpretation. This is the **diversity window** phenomenon. Consider the ELBO for the absorbing MDM:

$$\mathcal{L}(\theta) = \underbrace{\mathbb{E}_{t,x_0} [-\log p_\theta(x_0 | z_t)]}_{\text{reconstruction}} + \underbrace{\text{KL}(q(z_T) \| p(z_T))}_{\text{prior}} + \sum_{t=2}^T \underbrace{\text{KL}(q(z_{t-1} | z_t, x_0) \| p_\theta(z_{t-1} | z_t))}_{\text{transition KL}}$$

At low T (e.g. $T=128$), the reverse process must undo masking over fewer steps. Each step must unmask many tokens simultaneously, amplifying E_{fact} : the model’s conditional independence assumption introduces more error when more tokens are jointly unmasked per step. The resulting text is lower-quality but still diverse (the model hasn’t committed to any specific patterns, so the output varies).

At high T (e.g. $T=1000$), E_{fact} per step is small (few tokens unmasked per step), so local coherence improves. However, without remasking, once a token is committed in step t , it conditions all subsequent predictions. MDLM’s forward kernel is **non-remasking**: a token, once predicted, is never reconsidered. Over many steps, the sequence accumulates **commitment errors**: early tokens (predicted when context is sparse) condition later predictions, and the final sequence lies in a restricted subspace of the reference distribution $\rightarrow$ lower MAUVE.

At $T=256$, the model sits at the empirical optimum: E_{fact} is small enough for local coherence, but the number of steps is not so large that commitment errors accumulate. This suggests a **U-shaped quality curve** as a function of T for non-remasking models, with the minimum of `gen_ppl` and maximum of MAUVE at different optimal T .

Why remasking shifts this optimum. Remasking breaks the commitment error accumulation. For `remdm-loop`, MAUVE improves monotonically because the loop re-evaluates all committed

tokens at every step, preventing error accumulation. The “diversity window” only appears for MDLM because it is the only strategy without error correction.

3.4 Finding 3: Entropy Drop as a Signature of Mode Collapse

Entropy decreases for all strategies as T increases, but remdm-conf shows the steepest drop (5.499 $\rightarrow$ 5.357, $\Delta = 0.142$ bits over T=128 $\rightarrow$ 1000 vs 0.061 for mdlm).

Statistical interpretation. The entropy of the marginal character distribution is related to the **effective character diversity** of the generated texts. A useful summary statistic is the **character perplexity** 2^H , which equals the number of equally-probable characters needed to produce the same entropy — the information-theoretic analogue of vocabulary coverage:

$$2^{5.499} \approx 45.2 \quad (\text{remdm-conf, T=128}) \quad \rightarrow \quad 2^{5.357} \approx 40.8 \quad (\text{T=1000})$$

This represents a 9.7% reduction in character perplexity, indicating that at T=1000 the model’s output is drawn from a distribution equivalent to approximately 4 fewer effective characters. Note that this is the perplexity of the *marginal* character distribution — it does not capture reductions in word-level or sentence-level diversity. It provides a lower bound on the diversity reduction: actual semantic diversity could have decreased by more.

The combination of (a) the largest entropy drop, (b) the worst MAUVE at T=1000, and (c) the best gen_ppl at T=1000 for remdm-conf forms a **coherent statistical portrait of mode collapse**: - The model generates a narrower set of characters/words (entropy drop). - Those words are locally coherent and GPT-2-like (low gen_ppl). - But the global distribution does not match the reference (low MAUVE).

This is the **mode-seeking vs mode-covering** trade-off studied in the GAN literature (Theis et al., 2016): confidence-based remasking incentivises mode-seeking (commit the most likely token at each step), while loop remasking is more mode-covering (revisit all positions, allowing less likely but globally coherent trajectories).

3.5 Finding 4: gen_ppl Anomaly for MDLM at T=512 $\rightarrow$ T=1000

MDLM gen_ppl worsens slightly from T=512 (49.019) to T=1000 (52.269), while both remasking strategies continue to improve. This is anomalous: more denoising steps should only help (lower E_{fact} per step).

Hypotheses: 1. **Statistical noise.** With N=100 samples, the standard error on gen_ppl is non-trivial. The difference (49.0 vs 52.3) may lie within 1–2 standard deviations. A proper significance test (paired Wilcoxon or bootstrap CI) is needed to confirm this is a real effect. 2. **Commitment error accumulation.** As argued in Finding 2, non-remasking MDLM accumulates commitment errors at high T. At T=1000 with many small unmasking steps, early committed tokens — predicted in low-context windows — may condition later tokens adversely. The resulting sequences may still be locally coherent but have unusual long-range structure that GPT-2 assigns lower probability to. 3. **Numerical instability in the SDPA fallback.** Our setup uses PyTorch SDPA (no flash_attn). Numerical differences from the reference implementation may be amplified over T=1000 forward passes in a way not present at T=512.

This anomaly deserves attention: if real, it suggests MDLM has a non-monotone quality curve in `gen_ppl` as well as MAUVE, with an optimal `T` around 512. This would be a notable finding contradicting the paper’s implication of monotone improvement.

3.6 Alignment with arXiv:2503.00307

Finding	Our result	Paper claim	Assessment
MDLM <code>gen_ppl</code> at <code>T=128</code>	60.914	61.5	Consistent (<1% diff)
ReMDM <code>gen_ppl</code> improvement direction	<code>conf</code> < <code>loop</code> < <code>mdlm</code> at <code>T 512</code>	ReMDM > MDLM	Consistent
Remasking helps MAUVE	at <code>T=1000</code> , <code>loop</code> wins	claimed	Consistent
MAUVE absolute values	~10× higher than paper	—	Reference mismatch (expected)
MDLM MAUVE peak at <code>T=256</code>	Novel	Not reported	Novel finding
<code>remdm-conf</code> diversity collapse	Novel	Not reported	Novel finding
<code>remdm-loop</code> monotonic MAUVE	0.396→0.684	Implied	Consistent
MDLM <code>gen_ppl</code> worsens <code>T=512→1000</code>	Anomalous	Not reported	Needs testing

Note on MAUVE absolute values. The paper’s MAUVE values are approximately 6–11× lower than ours. The most likely explanation is reference corpus differences: MAUVE is not an absolute metric, and values are only comparable within an experiment using the same reference, sample size, and implementation. Our values use OWT (`skip=5000`, `N=1000` reference samples), which is a well-matched reference for a model trained on OWT. The paper likely uses a different reference (possibly `wikitext-103` or a different OWT slice) and possibly different `N`.

However, reference corpus alone may not account for the entire 6–11× gap. Other contributing factors include: (a) different sequence lengths for generated samples, (b) different `N` for the generated set, (c) different MAUVE hyperparameters (number of GMM clusters `k`, `max_text_length`), (d) potentially different generation quality. **In the absence of the paper’s exact experimental configuration, no quantitative attribution can be made.** What can be stated: within our experiment, MAUVE rankings are consistent and the relative improvements are interpretable. Cross-paper absolute comparisons should be avoided.

4 Research Directions

The following six directions emerge directly from the theoretical gaps and empirical anomalies identified above. Each is stated as a formal research question with a proposed methodology and expected contribution.

4.1 Direction 1: Confidence Calibration and Temperature-Scaled Remasking

4.1.1 Motivation

The remdm-conf diversity collapse (Finding 1) is characteristically a **calibration failure**: the model’s confidence scores $c^i = \max_j p_\theta(x^i = j \mid x_t)$ become systematically overconfident as T increases. At high T , the context is rich (few masked positions), so the softmax distribution over x^i is sharply peaked. The model commits tokens aggressively, reducing diversity.

4.1.2 Research Question

Can temperature scaling of the confidence signal prevent diversity collapse without sacrificing gen_ppl improvement?

4.1.3 Formal Proposal

Define a temperature-scaled confidence:

$$c_\tau^i = \max_j \text{softmax} \left(\frac{\ell_j^i}{\tau(t)} \right)$$

where ℓ^i are the logits at position i and $\tau(t)$ is a time-dependent temperature. Choices of $\tau(t)$:

- **Fixed temperature:** $\tau(t) = \tau_0 > 1$ (uniformly soften predictions)
- **Annealed temperature:** $\tau(t) = 1 + (\tau_0 - 1) \cdot \frac{t}{T}$, which gives $\tau = \tau_0$ at $t = T$ (start, fully masked, high exploration) and $\tau \rightarrow 1$ at $t = 1$ (end, mostly unmasked, no softening). Note: in MDM sampling, t counts *down* from T to 1; high t corresponds to early generation (sparse context, high masking rate).
- **Entropy-adaptive:** $\tau(t)$ chosen so that $H(\text{softmax}(\ell^i/\tau)) = H_{\text{target}}(t)$ for a pre-specified target entropy schedule $H_{\text{target}}(t)$

The annealed variant is particularly motivated: early in generation (large t , high masking rate, sparse context) use high temperature (explore diverse completions); late in generation (small t , low masking rate, rich context) use low temperature (commit to the best tokens). This mirrors **simulated annealing** and has a clean interpretation as an entropy regulariser on the commitment policy.

Remasking policy with entropy constraint:

$$S_t^* = \arg \max_{S: |S|=k_t} \sum_{i \in S} c_\tau^i \quad \text{subject to} \quad H(p_\theta^S) \geq H_{\min}(t)$$

where $H_{\min}(t)$ is a target minimum diversity at step t .

4.1.4 Expected Contribution

- Empirical: show that temperature scaling restores MAUVE at high T without significantly worsening gen_ppl.
- Theoretical: connect to the calibration literature (Guo et al., 2017) and derive conditions under which calibrated confidence is sufficient for principled remasking (extending the thesis’s three-tier hierarchy).

4.1.5 Implementation

No retraining required. Modify `external/remdm/main.py` sampling loop: add `temperature` parameter to the `remdm-conf` strategy. Run step-sweep (T=128/256/512/1000) for `{1.0, 1.5, 2.0, 3.0}` and entropy-adaptive variant.

4.2 Direction 2: Theoretical Characterisation of the Diversity Window

4.2.1 Motivation

Finding 2 shows that MDLM MAUVE has a non-monotone shape in T, peaking at T=256. This is currently an empirical observation. A theoretical explanation would be a genuine contribution.

4.2.2 Research Question

Can we derive an analytical expression for the optimal step count T^* (maximising MAUVE or minimising KL to reference) for a non-remasking MDM?

4.2.3 Formal Proposal

Consider the absorbing MDM reverse process. At step t , the expected factorization error for a single unmasking step of k_t tokens is:

$$E_{\text{fact}}(k_t) = \mathbb{E}_{x_0} \left[\text{KL} \left(p(x_{\mathcal{M}_t} \mid x_{\mathcal{M}_t}) \parallel \prod_{i \in \mathcal{M}_t} p_\theta(x^i \mid x_{\mathcal{M}_t}) \right) \right]$$

For a masking schedule where $k_t = n/T$ tokens are unmasked per step (uniform), E_{fact} decreases as T increases (fewer tokens unmasked per step, smaller approximation error). Define $E_{\text{fact}}^{\text{total}}(T) = \sum_{t=1}^T E_{\text{fact}}(k_t)$.

However, for **non-remasking** MDMs, there is a **commitment accumulation** term: tokens committed at step t with errors then condition all future predictions. Define the propagated error as:

$$E_{\text{commit}}(T) = \mathbb{E} \left[\sum_{t=1}^T \sum_{i \text{ committed at } t} \epsilon_t^i \cdot \Delta_t^i \right]$$

where ϵ_t^i is the prediction error at position i step t , and Δ_t^i is the downstream impact on future predictions (a measure of how strongly x^i conditions the remaining masked positions). As T increases,

ϵ_t^i decreases (more context) but the early-committed tokens (predicted when context is sparse) can have large Δ_t^i .

The hypothesis is that quality $\propto -(E_{\text{fact}}^{\text{total}}(T) + E_{\text{commit}}(T))$ has a maximum at finite T^* , giving the diversity window. Establishing this would require:

1. A tractable model of $E_{\text{commit}}(T)$ — potentially using Gaussian approximations or small-vocabulary toy models.
2. Verification that the model predicts $T^* \approx 256$ for MDLM on OWT.
3. A prediction for how remasking (loop or confidence) modifies T^* — the hypothesis is that remasking effectively suppresses E_{commit} , making T^* much larger or eliminating the diversity window entirely.

4.2.4 Expected Contribution

A theoretical bound on T^* as a function of model capacity, vocabulary size, and masking schedule. This would directly inform practitioners on how many steps to use and why.

4.3 Direction 3: Joint Remasking Policies

4.3.1 Motivation

Current remasking strategies treat positions **independently**: the decision of whether to remask position i depends only on c^i , not on the joint relationship between positions. The factorization error E_{fact} arises precisely because tokens are jointly dependent but treated independently. Remasking targeting **groups** of dependent tokens could reduce E_{fact} more efficiently.

4.3.2 Research Question

Can we design a polynomial-time remasking policy that selects the optimal set S^* of positions to remask jointly, by modelling pairwise or higher-order dependencies?

4.3.3 Formal Proposal

Define the **mutual information gain** from remasking a set S :

$$\text{MIG}(S) = I(x_S; x_{\setminus S}) = H(x_S \mid x_{\setminus S}) - H(x_S \mid x)$$

Remasking S is most beneficial when $\text{MIG}(S)$ is large: the positions in S are highly uncertain given the current context but would be better predicted jointly. The optimal remasking set is:

$$S^* = \arg \max_{S: |S|=k} \text{MIG}(S)$$

This is NP-hard in general (exponential search over subsets). However, $\text{MIG}(S)$ is a **submodular function** (by the submodularity of entropy), so a greedy algorithm achieves a $(1 - 1/e)$ approximation guarantee (Nemhauser et al., 1978).

Practical approximation. Compute pairwise attention weights a_{ij} from the transformer’s attention matrix as a proxy for mutual information. Define a graph G where edge (i, j) has weight

a_{ij} . Apply graph clustering (e.g. spectral clustering with k clusters) to identify groups of highly correlated positions. Remask the lowest-confidence member of each cluster.

4.3.4 Expected Contribution

- A theoretically grounded remasking policy that accounts for token dependencies.
- If the greedy MIG policy outperforms position-independent remasking, it provides evidence that token dependencies are the key bottleneck in current MDMs.
- A formal connection between remasking and the submodular optimisation literature.

4.3.5 Note on Difficulty

Estimating $\text{MIG}(S)$ from attention weights is an approximation whose quality is unknown. A cleaner (but more expensive) approach uses the model’s own forward pass to estimate $H(x^i \mid x_S \cup x_{(S \cup \{i\})})$ by running the model with different masked subsets — feasible only with small $|S|$.

4.4 Direction 4: Statistical Testing Framework for MDM Evaluation

4.4.1 Motivation

All papers in the field (including ours) report point estimates with no confidence intervals. At $N=100$, MAUVE estimates have substantial variance. The $T=512$ dip for remdm-loop ($0.614 \rightarrow 0.532$ before recovering to 0.684 at $T=1000$) may be noise. The MDLM gen_ppl anomaly ($49.0 \rightarrow 52.3$) may be noise. **Without proper statistical testing, these findings cannot be distinguished from random variation.**

4.4.2 Research Question

What is the minimum sample size N^* for statistically reliable MDM evaluation, and which differences in our results are statistically significant?

4.4.3 Formal Proposal

Bootstrap confidence intervals for MAUVE. Generate $B = 1000$ bootstrap resamples of the generated texts (with replacement, size N). For each resample, recompute MAUVE. The 95% CI is the 2.5th and 97.5th percentiles of the bootstrap distribution. A difference between two strategies is significant if their CIs do not overlap.

Paired test for gen_ppl. Per-sequence cross-entropies are paired (same seed gives comparable samples). Use a paired Wilcoxon signed-rank test (non-parametric, no Gaussian assumption) to test H_0 : median gen_ppl is equal between strategies.

Minimum N analysis. Using the bootstrap CIs as a function of N :

$$N^*(\delta) = \min\{N : \text{CI width} < \delta\}$$

for a target precision δ (e.g. $\delta = 0.05$ for MAUVE on $[0,1]$ scale). Our pilot data ($N=100$) can seed this analysis: fit the CI width as a function of $\sqrt{N}$ (standard error scaling) to extrapolate to larger N .

Implementation. Resample from the already-generated texts in `results/*/generated_sequences.json` — no additional HPC compute required for the bootstrap. Write `scripts/bootstrap_ci.py`.

4.4.4 Expected Contribution

The first systematic analysis of statistical reliability in MDM evaluation. The finding that $N=100$ is insufficient for reliable MAUVE (if demonstrated) would be a methodological contribution to the field. Conversely, showing that our key findings survive proper testing would strengthen the thesis’s empirical claims.

4.5 Direction 5: Adaptive Step Allocation

4.5.1 Motivation

All our evaluations use a **uniform step budget**: T total steps, each unmasking n/T tokens. But tokens differ in difficulty: function words (“the”, “of”) are easy to predict from context; content words in idiomatic phrases are hard. An adaptive policy that allocates more steps (smaller unmasking batches) to hard positions and fewer to easy ones could achieve the same quality as high- T uniform with fewer total forward passes.

4.5.2 Research Question

Can an adaptive step allocation policy match $T=1000$ quality with $T \approx 256$ forward passes, by concentrating compute on high-uncertainty positions?

4.5.3 Formal Proposal

Define token difficulty at step t as:

$$d_t^i = H(p_\theta(x^i \mid x_{\mathcal{M}_t})) = - \sum_j p_\theta^{ij} \log p_\theta^{ij}$$

(conditional entropy, not max-probability). Note this differs from the confidence signal $c^i = \max_j p_\theta^{ij}$: entropy captures full distributional uncertainty, not just the top probability.

Adaptive policy: 1. At each step, compute d_t^i for all $i \in \mathcal{M}_t$. 2. Set unmasking batch size k_t adaptively: $k_t \propto (1 - \bar{d}_t / \log_2 V)$, where $\bar{d}_t$ is the mean entropy over masked positions and V is vocab size. When average uncertainty is high (early steps), unmask fewer tokens. When uncertainty is low (late steps), unmask many tokens. 3. Always unmask the lowest-entropy (most confident) positions first.

This policy reduces to standard uniform unmasking when all entropies are equal. The step count T becomes a budget parameter, not a fixed schedule.

Connection to EB-Sampler. The EB-Sampler paper (Part IV of the literature review) derives the information profile $I^i(x)$ that optimally schedules when each position should be unmasked. Adaptive step allocation is an inference-time approximation to the EB-Sampler’s optimal schedule without requiring explicit estimation of I^i .

4.5.4 Expected Contribution

If adaptive allocation achieves $T=1000$ quality at fewer steps, it has direct practical value: faster inference at the same quality. It also provides an empirical test of the EB-Sampler’s theoretical prediction that step timing matters.

4.6 Direction 6: MAUVE as a Fine-Tuning Signal

4.6.1 Motivation

Our results show that MAUVE and `gen_ppl` are often in tension: the strategy that maximises `gen_ppl` (remdm-conf) minimises MAUVE at high T . If the goal is to generate text that matches a reference distribution (e.g. for data augmentation, style transfer), directly optimising for MAUVE during fine-tuning would align training and evaluation.

4.6.2 Research Question

Can a differentiable approximation of MAUVE be used as a reward signal for RL fine-tuning of a masked diffusion model, and does it improve MAUVE without collapsing `gen_ppl`?

4.6.3 Formal Proposal

MAUVE is not directly differentiable (it uses a discrete GMM fit in embedding space). However, a **soft MAUVE surrogate** can be constructed:

$$\widetilde{\text{MAUVE}}(p_\theta, q_{\text{ref}}) = -\text{KL}(f_\phi(p_\theta) \| f_\phi(q_{\text{ref}}))$$

where f_ϕ is a fixed feature extractor (e.g. frozen GPT-2 last hidden state). The gradient $\nabla_\theta \widetilde{\text{MAUVE}}$ can be computed via REINFORCE or via the reparameterisation trick if a differentiable sampler is available.

Connection to RemeDi. This is essentially the objective that RemeDi-RL was trained on (RL fine-tuning of a masked diffusion model for text quality). However, RemeDi-RL uses human feedback as the reward, not MAUVE. Using MAUVE as reward is cheaper (no human annotation) and more directly connected to the evaluation metric.

4.6.4 Expected Contribution

Demonstrates a principled training objective that directly targets distributional similarity. If successful, this closes the loop between evaluation and training for MDMs — an important step toward reliable MDM development. This direction also contextualises RemeDi’s RL approach: RemeDi is an instance of this broader paradigm.

4.7 Summary of Research Directions

Direction	Difficulty	Training required	Key contribution
1. Temperature-scaled remasking	Low	No	Prevents diversity collapse, no new training
2. Diversity window theory	Medium	No	Formal explanation of T^* optimum
3. Joint remasking policies	High	No	Dependency-aware remasking, submodular connection
4. Statistical testing framework	Low	No	Methodological contribution, N^* for MAUVE
5. Adaptive step allocation	Medium	No	Faster inference at same quality
6. MAUVE fine-tuning	High	Yes	Aligns training and evaluation objectives

Recommended next step: Direction 4 (statistical testing) can be implemented immediately with existing data (`results/*/generated_sequences.json`) and would validate or refute the anomalous findings (MDLM gen_ppl at $T=1000$, remdm-loop MAUVE dip at $T=512$) before investing in more experiments.

Strongest thesis contribution: Direction 1 (temperature scaling) + Direction 2 (diversity window theory) form a coherent story: empirically show that calibrated confidence prevents the collapse (Direction 1), theoretically explain why overconfidence causes it (Direction 2). Both can be done without retraining.

5 Appendix: Connection Map Between Papers and Findings

Paper	Key concept used in analysis
D3PM (Austin et al., 2021)	Absorbing Markov chain, forward/reverse process definition
MDLM (Sahoo et al., 2024)	ELBO simplification, factorization error formulation
SEDD (Lou et al., 2024)	Score-based perspective; connects <code>gen_ppl</code> to score matching
Lavenant & Zanella (2024)	KL error decomposition $\rightarrow$ formal E_{fact} definition
EB-Sampler	Optimal unmasking schedule $\rightarrow$ connection to Direction 5
ReMDM (arXiv:2503.00307)	Baseline for <code>gen_ppl</code> comparison; confidence/loop strategies
Mask-Predict (2019)	Historical precedent for confidence-guided remasking
MAUVE (Pillutla et al., 2021)	Formal definition of the distributional metric

All papers covered in `docs/comparison.md`. The above table maps findings to their theoretical grounding to aid thesis chapter writing.